

A SHARP DICHOTOMY FOR BOUNDED-GAP SUMSETS

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ABSTRACT. Let $k \geq 3$ and $1 \leq d_1 < d_2$. We prove that Erdős Problem 1112 has an affirmative answer exactly when $d_2 \geq k + 1$; in that range the explicit lacunarity ratio $d_2 + 2$ works. Below the threshold we prove more: however rapidly prescribed successive ratios grow, one sequence meets the k -fold sumset of every admissible bounded-gap sequence. The proof combines a reciprocal Beatty construction, Kneser's density theorem, and an elementary proof of the sharp finite subset-sum lemma required at the boundary.

1. INTRODUCTION

For a set $A \subset \mathbb{N}$, write

$$kA := \{a_1 + \cdots + a_k : a_1, \dots, a_k \in A\},$$

with repetitions allowed. A sequence $A = \{a_1 < a_2 < \cdots\}$ is (d_1, d_2) -admissible if $d_1 \leq a_{n+1} - a_n \leq d_2$ for every n . We consider the following formulation recorded as Erdős Problem 1112 [1], inspired by a remark of Erdős and Graham [2].

Problem. Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers. Does there exist an integer r such that every increasing sequence $B = \{b_1 < b_2 < \cdots\}$ of positive integers satisfying $b_{n+1} \geq rb_n$ admits an increasing sequence $A = \{a_1 < a_2 < \cdots\}$ of positive integers with

$$d_1 \leq a_{n+1} - a_n \leq d_2 \quad (n \geq 1), \quad kA \cap B = \emptyset?$$

The following theorem determines exactly when such a ratio exists.

Theorem 1 (Sharp dichotomy). *Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers.*

- (1) *If $d_2 \geq k + 1$, every sequence of positive integers $B = \{b_1 < b_2 < \cdots\}$ with $b_{n+1} \geq (d_2 + 2)b_n$ admits a (d_1, d_2) -admissible A such that $kA \cap B = \emptyset$.*
- (2) *If $d_2 \leq k$, no fixed ratio works. More strongly, for every sequence $R_1, R_2, \dots$ of positive integer lower bounds there is one increasing sequence B of positive integers with $b_{n+1} \geq R_n b_n$ which meets kA for every (d_1, d_2) -admissible A .*

Thus the answer is yes exactly when $d_2 \geq k + 1$.

The earlier proof [5], formally verified in Lean, established this dichotomy with the bound $192d_2$. The present proof improves that bound and replaces the finite certificate argument with an elementary construction.

Above the threshold, a reciprocal Beatty sequence avoids a prescribed lacunary sequence. Below it, every admissible walk has a congruence-class tail in its k -fold sumset; a diagonal construction then meets every such sumset. The finite interval lemma, Theorem 3.2, supplies the required offsets at the boundary. Its proof is given in Section 6. Appendix A records the correspondence with the Lean formalization.

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2. THE EXISTENCE CONSTRUCTION

Put $D = d_2$ and assume $D \geq k + 1$.

Lemma 2.1 (Lacunary interpolation). *Let $I = [u, v] \subset (0, 1)$ have length $\ell > 0$, let J be a nondegenerate closed bounded real interval, and let positive integers $q_1 < q_2 < \dots$ satisfy $q_{n+1} \geq \rho q_n$. If*

$$\ell(\rho - 1) \geq 1, \quad (1)$$

then, after deleting finitely many q_n , some $\theta \in J$ satisfies $\{q_n \theta\} \in I$ for every remaining n .

Proof. For a positive integer q , the set

$$E_q = \{\theta : \{q\theta\} \in I\} = \bigcup_{m \in \mathbb{Z}} [(m+u)/q, (m+v)/q]$$

has components of length ℓ/q , with left endpoints spaced by $1/q$. For some N , the interval $[q_N \inf J - u, q_N \sup J - v]$ has length $q_N |J| - \ell \geq 1$, so it contains an integer. Thus a component K_N of E_{q_N} lies in J . If K_n is a component of E_{q_n} , the allowable indices of components of $E_{q_{n+1}}$ contained in K_n form a real interval of length

$$q_{n+1}|K_n| - \ell = \ell(q_{n+1}/q_n - 1) \geq 1.$$

It contains an integer, giving $K_{n+1} \subseteq K_n$. The nested nonempty compact intervals have a common point θ , as required. $\square$

Proposition 2.2 (Existence). *If $D \geq k + 1$, the ratio $D + 2$ works.*

Proof. Given B with $b_{n+1} \geq (D + 2)b_n$, put

$$\varepsilon = \frac{1}{2D(D+1)}, \quad I = [\varepsilon, \varepsilon + 1/(D+1)], \quad J = [1/D, 1/D + \varepsilon/(D-1)].$$

Lemma 2.1 applies with equality in (1). It yields $\theta \in J$ and N_0 such that

$$\{b_n \theta\} \in I \quad (n \geq N_0). \quad (2)$$

Put $\alpha = 1/\theta$. Then

$$D - 1 < \alpha \leq D, \quad (3)$$

and, since $k \leq D - 1$,

$$1 - k\theta \geq 1 - (D - 1) \max J = 1/D - \varepsilon = \varepsilon + 1/(D + 1) = \max I. \quad (4)$$

Choose an integer $m \geq 1$ with $k \lfloor \alpha m \rfloor > \max_{n < N_0} b_n$, reading the empty maximum as zero, and set

$$A = \{\lfloor \alpha(m + i) \rfloor : i = 0, 1, 2, \dots\}.$$

Its gaps are $D - 1$ or D , hence admissible, and its least k -fold sum exceeds every discarded term. If a remaining b were a sum of k such floors, then for an integer S (the sum of their indices) $b \leq \alpha S < b + k$, whence

$$b\theta \leq S < (b + k)\theta.$$

But $0 < \{b\theta\} \leq 1 - k\theta$ by (2)–(4), so this half-open interval contains no integer. This is a contradiction. $\square$

Remark 2.3. The argument works for every real ratio $\rho > 1 + D/(D - k)$: take J sufficiently close to $1/D$ and a closed interval $I \subset (0, 1 - k \sup J)$ of length $1/(\rho - 1)$.

3. TOOLS FOR TAIL COVERING

Call A *k-tail-covering* if, for integers $q \geq 1$, $0 \leq r < q$, and X_0 ,

$$\{x \geq X_0 : x \equiv r \pmod{q}\} \subseteq kA. \quad (5)$$

3.1. Density and finite subset sums. For $X \subseteq \mathbb{N}_0$ write $\underline{d}(X) = \liminf_{N \rightarrow \infty} |X \cap [1, N]|/N$.

Lemma 3.1 (A consequence of Kneser's theorem). *Let $P = \{p_0 < p_1 < \dots\}$ with $p_0 = 0$. If $p_n \leq cn + C$ eventually, for constants $0 < c < k$ and C , then kP contains a congruence-class tail.*

Proof. For large N , take $n = \lfloor (N - C)/c \rfloor$. Then $p_n \leq N$, so $\underline{d}(P) \geq 1/c > 1/k$. The lower-density consequence of Kneser's theorem [4, 3] needed here is that either $\underline{d}(kP) \geq k\underline{d}(P)$ or kP contains a congruence-class tail. The first alternative is impossible because density is at most one, proving the claim. $\square$

For a finite multiset S , write $\text{SS}(S) = \{\sum_{x \in T} x : T \text{ is a submultiset of } S\}$.

Theorem 3.2 (Finite interval lemma). *If G is a finite set of positive integers with $|G| \geq 3$, $\gcd(G) = 1$, and $M = \max G$, some multiset of at most $M - 1$ elements of G has M consecutive subset sums.*

We prove this in Section 6. First we explain how an interval of subset sums yields representations of all sufficiently large integers.

3.2. A slot lemma. For a gap word $g_1, g_2, \dots$, put $p_0 = 0$ and $p_n = g_1 + \dots + g_n$. A tail of the walk is a translate of $P = \{p_n : n \geq 0\}$.

Lemma 3.3 (Slots). *Suppose all gaps belong to a finite G , every member recurs infinitely often, $|G| \geq 3$, $\gcd(G) = 1$, and $M = \max G \leq k$. Then kP contains every sufficiently large integer.*

Proof. By Theorem 3.2 choose $S = \{\delta_1, \dots, \delta_m\}$, $m \leq M - 1 \leq k - 1$, with $[c, c + M - 1] \subseteq \text{SS}(S)$. Choose distinct occurrences $g_{n_j} = \delta_j$. In summand slot j choose p_{n_j-1} or p_{n_j} . These slots realize a fixed base plus every offset in $[c, c + M - 1]$. Set another $k - 1 - m$ summands equal to $p_0 = 0$ and let the last summand vary over p_n . For a large target x , the residual $x - \text{base} - p_n$ decreases to $-\infty$ in jumps at most M . Its first value at most $c + M - 1$ is at least c , so the slots supply the residual. This represents x in kP . $\square$

4. EVERY SUBCRITICAL WALK IS TAIL-COVERING

Proposition 4.1. *For $k \geq 3$, every increasing sequence of positive integers with gaps at most k is k -tail-covering.*

Proof. Discard a finite prefix so that the gaps lie in the nonempty tail alphabet G of values recurring infinitely often. Put $p_0 = 0$ and $p_n = g_1 + \dots + g_n$. Divide all gaps and prefix sums by $g = \gcd(G)$. A class tail modulo q rescales to one modulo gq ; translating back only changes the residue. Relabel so that $\gcd(G) = 1$. If $M = \max G < k$, then $p_n \leq Mn$, and Lemma 3.1 applies. Thus assume $M = k$. If $|G| \geq 3$, use Lemma 3.3. It remains to consider an alphabet with two elements: a singleton would have $\gcd k > 1$. Hence $G = \{\delta, k\}$ with $1 \leq \delta < k$ and $\gcd(\delta, k) = 1$. Set $e = k - \delta$, encode gap k by $h_n = 1$ and gap δ by $h_n = 0$, and write $q_n = h_1 + \dots + h_n$. Then

$$p_n = \delta n + eq_n. \tag{6}$$

For $s \geq 0$ set

$$W_s = \left\{ \sum_{j=1}^k q_{i_j} : i_j \geq 0, \sum_{j=1}^k i_j = s \right\}.$$

This is an integer interval $[w^-(s), w^+(s)] \cap \mathbb{Z}$: compositions are connected by unit transfers, each changing the value by $-1, 0$, or 1 . A path between extremizers misses no intermediate value. Adding or deleting one unit shows that each endpoint rises by 0 or 1 as s increases by one. Write $\text{width}(s) = w^+(s) - w^-(s)$.

Sweep. If $\text{width}(s) \geq k - 1$ eventually, then kP contains every large integer. Indeed, a target x is represented at budget s exactly when $w(s) = (x - \delta s)/e \in W_s$. The integral values of $w(s)$ occur in one class of s modulo e . Along this class w drops by δ per step, whereas either endpoint rises by at most e . Since $w^+(s) \leq s$, for large x the target starts above the band at the first admissible budget

beyond the width threshold; eventually it becomes negative. If it skips the band, consecutive admissible budgets satisfy

$$w^+(s) < w(s) = w(s+e) + \delta \leq w^-(s+e) - 1 + \delta \leq w^-(s) + e - 1 + \delta.$$

Thus $\text{width}(s) \leq k-2$, a contradiction.

A binary word is *balanced* if equal-length factors differ in their numbers of ones by at most one. Define

$$V(\sigma) = \max_{i+j=\sigma} (q_i + q_j) - \min_{i+j=\sigma} (q_i + q_j).$$

If $V(\sigma) \leq 1$ for every σ , the word is balanced: comparing factors with endpoint indices i, i' and j', j , where $i+j = i'+j'$, bounds their count difference by $V(i+j)$. Let $z_n = n - q_n$ count zeros. Then

$$p_n = kn - ez_n. \tag{7}$$

If $\liminf z_n/n > 0$, choose $0 < \gamma \leq 1$ with $z_n \geq \gamma n$ eventually; then $p_n \leq (k - e\gamma)n$, and Lemma 3.1 applies. Assume $\liminf z_n/n = 0$. There are infinitely many σ with $V(\sigma) \geq 2$. Otherwise a sufficiently late tail would be balanced by the same endpoint comparison. Since zero recurs, it has a length- L factor containing two zeros. Every length- L factor would then contain a zero, giving $z_n \geq n/L - O(1)$, a contradiction.

If $k = 2u + 1$, fix σ_0 with $V(\sigma_0) \geq 2$. For all large s , split the slots into u pairs of subtotal σ_0 and a singleton carrying the rest. Independent variation gives width at least $uV(\sigma_0) \geq k-1$, so the sweep applies.

For even k , pairing all the summands leaves no singleton to absorb the remaining index sum. We instead show that repeated failures of the width bound would force the binary word to be periodic. Write $k = 2u$, $u \geq 2$, and choose $\sigma_0 < \sigma_1$ with $V(\sigma_i) \geq 2$, and put $\Delta = \sigma_1 - \sigma_0$. Suppose at a sufficiently large s the width is at most $k-2$, and put $\tau = s - (u-1)\sigma_0$. Using $u-1$ pairs of subtotal σ_0 and one of subtotal τ gives width at least $k-2 + V(\tau)$, so $V(\tau) = 0$. Using $u-2$ pairs of subtotal σ_0 , one of subtotal σ_1 , and one of subtotal $\tau - \Delta$ likewise gives $V(\tau - \Delta) = 0$. Now $V(n) = 0$ implies $q_i + q_{n-i}$ is constant. Successive differences give $h_{i+1} = h_{n-i}$, so the prefix of length n is a palindrome. The prefixes of lengths τ and $\tau - \Delta$ are therefore palindromes; the shorter is also a suffix of the longer, which consequently has period Δ . Infinitely many width failures would give such periodic prefixes of unbounded length, forcing the whole word to have period Δ . A period contains a zero, contradicting $\liminf z_n/n = 0$. Thus width is eventually at least $k-1$, and the sweep finishes. $\square$

5. PROOF OF THE NON-EXISTENCE DIRECTION

The preceding proposition gives a congruence-class tail for each admissible sequence. The following diagonal construction meets all such tails while allowing arbitrary prescribed growth.

Lemma 5.1. *If every admissible walk is k -tail-covering, then any prescribed positive integer ratios R_n admit one increasing B with $b_{n+1} \geq R_n b_n$ meeting kA for every admissible A .*

Proof. List pairs (q, r) , $q \geq 1$, $0 \leq r < q$, with each recurring infinitely often, by concatenating the finite lists with $q \leq N$ for $N = 1, 2, \dots$. Write the list as (q_n, r_n) . Choose $b_1 > 0$ in its prescribed class and let b_{n+1} be the least integer in the next class exceeding $\max\{R_n b_n, b_n, n\}$. The sequence grows as required. For any A , its class in (5) occurs infinitely often; the corresponding b_n are unbounded, so one exceeds X_0 and belongs to kA . $\square$

Proof of Theorem 1. Part (1) is Proposition 2.2. For $d_2 \leq k$, Proposition 4.1 and Lemma 5.1 give part (2). $\square$

6. PROOF OF THE FINITE INTERVAL LEMMA

We first reduce Theorem 3.2 to triples of pairwise coprime integers. Two constructions then cover the remaining triples, according to whether their two largest elements sum to a multiple of the smallest. Write u^m for the multiset of m copies of the positive integer u .

6.1. Reduction to three generators.

Lemma 6.1 (Coprime rectangle). *If $1 \leq p < q$ and $\gcd(p, q) = 1$, then*

$$\text{SS}(p^{q-1}, q^{p-1}) \supseteq [(p-1)(q-1), pq-1].$$

The interval has $p+q-1$ terms, at multiset cost $p+q-2$.

Proof. For n in the interval choose $0 \leq i < q$ with $ip \equiv n \pmod{q}$. Then $j = (n - ip)/q$ satisfies $-1 < j < p$, so $0 \leq j \leq p-1$. $\square$

Lemma 6.2 (Interlacing). *Let $d, q \geq 1$, $\gcd(d, q) = 1$, and $q \leq N$. If $[A, A+N-1] \subseteq \text{SS}(C)$, then scaling C by d and adjoining $d-1$ copies of q gives N consecutive subset sums.*

Proof. For $0 \leq n < N$ choose $0 \leq j < d$ with , and put $t = ((d-1)q + n - jq)/d$. Since $0 \leq (d-1)q + n - jq < dN$, we have $0 \leq t < N$. Thus

$$d(A+t) + jq = dA + (d-1)q + n$$

supplies the required interval. $\square$

Lemma 6.3 (Two-generator targets). *For coprime $1 \leq p < h$ and $N \geq 2h$, at most $N - \lfloor N/h \rfloor$ terms from $\{p, h\}$ have N consecutive subset sums.*

Proof. Start with the rectangle's interval of length $p+h-1$. Write $N = sh + r$, $s \geq 2$, $0 \leq r < h$, and add $t = s-1 + \mathbf{1}_{\{r \geq p\}}$ copies of h . The translates overlap or abut and cover at least N integers, at cost $h+p+s-3 + \mathbf{1}_{\{r \geq p\}}$. At $s=2$ its slack below $N-s$ is $h+r-p-1 - \mathbf{1}_{\{r \geq p\}} \geq 0$; increasing s increases slack by $h-2$. $\square$

Proposition 6.4 (Reduction to three generators). *Assuming Theorem 3.2 for maxima below M , it suffices at M to treat triples*

$$G = \{a < b < M\}, \quad \gcd(a, b) = \gcd(a, M) = \gcd(b, M) = 1, \quad a+b \geq M+2. \quad (8)$$

Proof. Set $H = G \setminus \{M\}$, $d = \gcd(H)$, and $e = \max(H/d)$. If $|H| \geq 3$, induction gives e consecutive subset sums at cost $e-1$ from H/d . Adjoin $\lceil M/e \rceil - 1$ copies of e , obtaining at least M consecutive sums. Scale by d and interlace with $d-1$ copies of M ; $\gcd(d, M) = 1$. The cost is $C = e + d + \lceil M/e \rceil - 3$. Writing $M = de + s$ with $s \geq 1$ gives

$$M - 1 - C = (d-1)(e-2) + s - \lceil s/e \rceil \geq 0,$$

since $e \geq 3$. Otherwise $H = \{a, b\}$, $a < b$. If $d = \gcd(a, b) > 1$, apply Lemma 6.3 to $\{a/d, b/d\}$ with target M , scale, and interlace with $d-1$ copies of M . Here $M > b$, $d \geq 2$, and $\lfloor M/(b/d) \rfloor \geq d$, so the cost is at most $M-1$. Now assume $\gcd(a, b) = 1$. If $b \leq M/2$, the same target lemma costs at most $M-2$. If $b > M/2$ and $a+b \leq M$, one more copy of b extends the rectangle to length $a+2b-1 \geq M$, at cost at most $M-1$. If $a+b = M+1$, the rectangle itself suffices. Thus $a+b \geq M+2$ remains. Finally, if $d = \gcd(a, M) > 1$, apply the target lemma to $\{a/d, M/d\}$ with target M , at cost at most $M-d$, scale, and interlace with $d-1$ copies of b ; $\gcd(d, b) = 1$. The case $\gcd(b, M) > 1$ is symmetric. $\square$

For a triple in (8), put

$$e = b - a, \quad h = M - b, \quad \mu = M - a = e + h. \quad (9)$$

Then

$$1 \leq h \leq a-2, \quad (10)$$

so $a \geq 3$ and $\gcd(a, e) = \gcd(a, \mu) = 1$.

6.2. Paired movers.

Lemma 6.5 (Paired movers). *If $3 \leq a < b < M$, $\gcd(a, b) = 1$, and $b+M = ra$, then for*

$$x = r-1, \quad y = \lceil (M-r)/2 \rceil, \quad z = \lfloor (M-r)/2 \rfloor,$$

$\text{SS}(a^x, b^y, M^z)$ contains M consecutive integers and $x+y+z = M-1$.

Proof. Here $r \geq 3$, and

$$M \geq a + r - 1, \quad M \leq a(r - 1) - 1 : \quad (11)$$

indeed $2M \geq ra + 1 \geq 2a + 2r - 2$ and $M = ra - b$. Put $p = \lfloor a/2 \rfloor$, $q = \lfloor (a-1)/2 \rfloor$, and $L = \max(pb, qM)$. Then $p+q = a-1$, $p \leq y$, and $q \leq z$. For $n \in [L, L+M-1]$ choose $w \in [0, a-1]$ with $wb \equiv n \pmod{a}$. If $w \leq p$, use w unpaired copies of b and divide $(n-wb)/a = i+r\kappa$, $0 \leq i < r$. Otherwise put $w' = a-w \leq q$, use w' copies of M , and divide $(n-w'M)/a = i+r\kappa$. The numerators are nonnegative, so $\kappa \geq 0$. Each of κ further pairs (b, M) contributes ra , and $i \leq x$. The copies are available provided

$$L + M - 1 + pM \leq a(ry + x), \quad (12)$$

$$L + M - 1 + qb \leq a(rz + x). \quad (13)$$

For example, in the first branch $a(i+r\kappa) = n-wb$ and $a(i+r(\kappa+w)) = n+wM$; these inequalities bound $\kappa \leq z$ and $\kappa + w \leq y$. The second branch is symmetric. If $L = qM$, the inequalities reduce to $aM - 1 \leq a(ry + x)$ and $qra + M - 1 \leq a(rz + x)$. The latter uses $q \leq z$ and $M - 1 \leq ax$; the former uses

$$ry + x \geq r(M - r)/2 + r - 1 \geq M,$$

since $M \geq a + r - 1 \geq r + 2$. If $L = pb$, the first inequality follows from $p \leq y$ and $M - 1 \leq ax$. The second is $(a-1)b + M - 1 \leq a(rz + x)$. Odd a gives $p = q$ and $qM > pb$, so take $a = 2v$, $v \geq 2$. Put $\tau = M - vr = vr - b \geq 1$. The slack is

$$D = 2vrz - 2(v-1)b - 2v + 1 = 2vr(z - v + 1) + 2(v-1)\tau - 2v + 1.$$

Here $z = \lfloor (r(v-1) + \tau)/2 \rfloor \geq v-1$. If $z \geq v$, then $D \geq 2vr - 1 > 0$. If $z = v-1$, then $\tau \neq 1$ (that would give $z \geq v$), so $\tau \geq 2$ and $D \geq 2v - 3 > 0$. Finally $x + y + z = M - 1$. $\square$

6.3. The symmetric residue path.

Lemma 6.6 (Centered frame). *If C has a subset sum in one common interval $[L, U]$ for each residue modulo a , put $x = \lceil (M - 1 + U - L)/a \rceil$. Then $\text{SS}(C \cup a^x)$ contains $[U, L + ax]$, with at least M integers.*

Proof. Subtract the chosen representative of $n \pmod{a}$ from $U \leq n \leq L + ax$. The remainder is a nonnegative multiple of a at most ax . $\square$

Lemma 6.7 (Symmetric η argument). *If $a < b < M$ satisfy (10), are pairwise coprime, and $a \nmid b + M$, at most $M - 1$ terms from $\{a, b, M\}$ have M consecutive subset sums.*

Proof. Let p, q be units modulo a , let $\eta \in [0, a-1]$ represent qp^{-1} , and put

$$t = \min(\eta, a - \eta), \quad Z = \lfloor (a-1)/t \rfloor, \quad r = a - 1 - tZ, \quad K = t - 1 + Z. \quad (14)$$

For $(p, q) = (b, M)$ we have $\eta \equiv \mu e^{-1} \pmod{a}$. The values $0, -1, 1$ would give respectively $a \mid M$, $a \mid b + M$, and $a \mid h$, so $2 \leq t \leq \lfloor a/2 \rfloor$. If $\eta = t$, write $0 \leq n < a$ as $n = tk + j$. The representatives $jp + kq$ lie in $[0, S]$, where

$$S = \max\{(t-1)p + (Z-1)q, rp + Zq\}. \quad (15)$$

If $\eta = a - t$, write the consecutive residues $t - a, \dots, t - 1$ as $n = j - tk$; representatives lie in $[0, S]$ with

$$S = (t-1)p + Zq. \quad (16)$$

All are subset sums of p^{t-1}, q^Z . The centered frame with $L = 0$ costs

$$\mathcal{B} = K + \lceil (M - 1 + S)/a \rceil. \quad (17)$$

We can also reverse (p, q) , since M is a unit modulo a . Always

$$K \leq \lceil a/2 \rceil. \quad (18)$$

Indeed, for $c = \lceil a/2 \rceil$ the concave function $t(c - t + 2)$ exceeds $a - 1$ at both endpoints of $[2, \lfloor a/2 \rfloor]$, so $\lfloor (a-1)/t \rfloor \leq c - t + 1$. Since $S \leq KM$, the budget holds whenever

$$(a - K - 1)M \geq a(K + 1) - 1. \quad (19)$$

For $a = 3$ all unit residues are already excluded. We now check the bound $\mathcal{B} \leq M - 1$ separately for odd and even a . In each case, (19) handles the smaller values of K ; only its largest possible value requires a more precise choice of representatives.

Odd $a = 2n + 1$. If $K \leq n$, the slack in (19) at $K = n$ is $n(\mu - 2) \geq 0$. Furthermore $t - 1 + \lfloor 2n/t \rfloor = n + 1$ forces $t \in \{2, n\}$, since $(t - 2)(t - n) < 0$ for $2 < t < n$. Only $\eta \in \{2, -2, n, n + 1\}$ remains. If $\eta = 2$, then $e - h = \lambda a$ for $\lambda \geq 0$ by (10). Here $Z = n$, $r = 0$, $S = nM$. With $\epsilon = 0$ for $h = 1$ and $\epsilon = 1$ for $h \geq 2$, evaluation gives

$$(M - 1) - \mathcal{B} = n\lambda + h - 2 - \epsilon. \quad (20)$$

This is nonnegative if $\lambda \geq 1$, or if $\lambda = 0$ and $e = h = g \geq 3$. For $g = 1$, take $n + 1$ copies of a , one of $a + 1$, and n of $a + 2$. At selection level $n + 1$, offsets fill $[0, a]$; at the next level offsets $[1, a]$ remain. The intervals about, of total length $2a + 1 \geq a + 2$. For $g = 2$, take $n + 3$ copies of a , one of $a + 2$, and n of $a + 4$. With $c = n + 1$, each level $\ell \in \{c, c + 1, c + 2\}$ contains $\ell a + 2[0, a]$. On $[(c + 1)a, (c + 3)a]$, the middle level supplies one parity and the other levels supply the other. The costs are respectively $a + 1$ and $a + 3$, as required.

If $\eta = -2$, reverse b, M ; the reversed ratio is n . For $n \geq 3$, the normalized residues $1, \dots, a$ have representatives

$$jM \ (1 \leq j \leq n - 1), \quad b + jM \ (0 \leq j \leq n - 1), \quad 2b, \quad M + 2b.$$

These use M^{n-1}, b^2 and lie in $[b, b + (n - 1)M]$. The centered frame costs $n + 1 + \lceil (nM - 1)/a \rceil \leq M - 1$, since the slack is

$$a(M - n - 2) - (nM - 1) = (n + 1)\mu - 2n \geq 2.$$

For $n = 2$, representatives $M, b, M + b, 2b, M + 2b$ lie in $[b, M + 2b]$. The cost condition is $2e + 3h - 9 \geq 0$. Since $h \equiv 2e \pmod{5}$, pairwise coprimality and $1 \leq h \leq 3$ exclude $e = 1, 2$, so it holds.

For $\eta = n$ or $n + 1$ with $n \geq 3$, use the forward path in the first case and reverse in the second, making its ratio 2. In either case $K = n + 1$ and $S \leq (n - 1)b + M$. The budget is at most $a + 1 + \lceil ((n + 1)e + 2h - 1)/a \rceil$, and its required slack is

$$ne + (a - 2)h - 2a + 1 \geq 0. \quad (21)$$

For $h \geq 2$ this is at least $n - 3$. For $h = 1$, the first orientation gives $a \mid 3e + 2$, so $e \geq 3$; the second gives $a \mid e + 2$, so $e \geq a - 2$. Both imply the inequality.

Even $a = 2n$. If $K \leq n - 1$, (19) holds, with slack $n\mu + 1$ at $K = n - 1$. For $a \geq 12$ and $3 \leq t \leq n - 2$, $t - 1 + \lfloor (2n - 1)/t \rfloor \leq n - 1$. Thus maximal $K = n$ requires $t \in \{2, n - 1, n\}$. As η is a unit, only even n and $\eta = n \pm 1$ remain. For $\eta = n - 1$, (15) gives $Z = 2$, $r = 1$ and $S = (n - 2)b + M$. The slack is

$$ne + 2(n - 1)h - 2n + 1 \geq n - 1. \quad (22)$$

For $\eta = n + 1$, $h \equiv ne \pmod{2n}$; e is odd, so (10) forces $h = n$. Now $S = (n - 2)b + 2M$ and the slack is

$$2n^2 - 7n + (n - 1)e + 1 \geq 2n(n - 3) > 0. \quad (23)$$

For $a = 4, 6$ there are no unit residues beyond ± 1 . For $a = 8$, $\eta = 3$ gives $S = b + 2M$ and slack $8(M - 5) - (3M + b - 1) = 4e + 5h - 7 > 0$. For $a = 8$, $\eta = 5$ forces $h = 4$; $S = 2b + 2M$ gives slack $8(M - 5) - (3M + 2b - 1) = 3e + 5h - 15 > 0$. For $a = 10$, reverse if necessary to make the ratio 3. The height is at most $2b + 2M$, and slack is $10(M - 6) - (3M + 2b - 1) = 5e + 7h - 9 > 0$. This proves the required bound in every case. $\square$

Proof of Theorem 3.2. Use strong induction on M , with the statement vacuous for $M < 3$. Proposition 6.4 leaves a triple satisfying (8). If $a \mid b + M$, use Lemma 6.5; otherwise use Lemma 6.7. $\square$

USE OF AI TOOLS

Generative AI tools contributed to the development of the original proof, the drafting and revision of the manuscript, and the construction and debugging of the Lean formalization.¹ The shortened presentation is based on a manuscript supplied by Stijn Cambie. GPT-6-Astra and Claude Sonnet 5 assisted with its revision and formalization. The formalization was checked by Lean’s kernel. The author takes responsibility for the mathematical claims and their presentation.

APPENDIX A. FORMAL VERIFICATION AND CORRESPONDENCE

Both directions, including the density dependency and finite interval lemma, are checked in Lean. Selected declarations are listed below; paths are relative to `lean/Erdos1112Proof/`.

Paper result	Lean declaration
Existence/Reciprocal.lean	
Lacunary interpolation	<code>reciprocal_interpolation</code>
Existence with ratio $d_2 + 2$	<code>existence_bound_reciprocal</code>
Short/Intervals.lean	
Coprime rectangle	<code>coprime_rectangle</code>
Interlacing	<code>interlacing</code>
Two-generator targets	<code>two_generator_target</code>
Centered frame	<code>centered_frame</code>
Short/Slots.lean	
Slots and final-summand sweep	<code>slots_from_run</code>
Short/Normalization.lean	
Tail alphabet and gcd normalization	<code>normalized_walk</code>
Short/Binary.lean	
Zero-liminf binary width argument	<code>binary_eventual_width</code>
Binary covering from zero liminf	<code>binary_tail_covering</code>
Short/Funnel.lean	
Reduction to three generators	<code>sharpAt_of_funnel</code>
Short/Sharp.lean	
Complete finite interval lemma	<code>sharp_all</code>
Short/KneserWeak.lean	
Pairwise density/AP-tail alternative	<code>weak_kneser</code>
Short/Main.lean	
Consequence of Kneser’s theorem	<code>density_shortcut</code>
Subcritical walk covering	<code>all_tailCovering</code>
Universal lacunary construction	<code>strong_nonexistence</code>

The canonical theorems in `Final.lean` use these proofs throughout. The Kneser consequence is proved using e-transforms, residue compression and finite Mann estimates; it is not assumed. The old proofs and certificate tables have been removed. The definitions in `lean/Erdos1112.lean` allow repeated summands, phrase gap bounds additively, and prove equivalence of natural and integer ratio formulations. The repository supplies build and axiom-audit commands.

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¹In the original development, Claude (Fable 5 and Opus 4.8) contributed to the mathematical arguments; GPT-5.5 and Gemini 3.1 assisted with advice and review.